Chapter 1

Proof by Tableau

> If controversies were to arise, there would be no more need of disputation between two philosophers than between two accountants. For it would suffice to take their pencils in their hands, and say to each other: Let us calculate.
>
> — Leibniz

An argument is valid when its conclusion cannot be false while its premises are true. For a few propositional letters the definition can be checked by brute force: list every way of making the letters true or false, and look for a row where every premise is true and the conclusion false. There is no such row for modus ponens, from $p \to q$ and p to q:

p	q	$p \to q$	p	q
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

The table doubles with every letter, and says nothing about necessity or obligation, where a sentence's truth depends on other worlds than this one. A tableau, or truth tree, does better. Suppose the premises true and the conclusion false, and take the formulas apart by their connectives: $A \wedge B$ puts both A and B on the branch, while $A \vee B$ and $A \to B$ split it in two. A branch holding a formula and its negation is closed (×):

no way of making things true lies along it. If every branch closes, the supposition was impossible and the argument is valid. A branch that stays open describes a counterexample.

1.1 Modus ponens

Here is modus ponens again, as a tableau, with a third premise, that it is cold, which the proof never uses. Denying the conclusion (line 3) and splitting the conditional (line 1) closes both branches at once (figure 1.1).

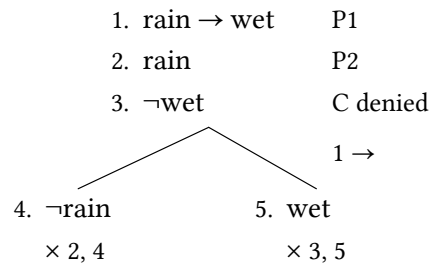


Figure 1.1. Modus ponens, by tableau: both branches close. The premise that it is cold (P3) is never used.

1.2 Necessity

Modal logic reads $\Box A$ as "necessarily A" and $\Diamond A$ as "possibly A". Following Kripke, a model is a set of worlds and a relation R of accessibility between them: $\Box A$ is true at a world when A is true at every world it sees, and $\Diamond A$ when A is true at some world it sees. Each formula in a tableau is now true at a world, w_0 to begin with. A $\Diamond$ opens a new world, and a $\Box$ sends its formula along every arrow already drawn. The axiom of the weakest normal logic, K, is that necessity distributes over the conditional:

$$\Box(p \to q),\ \Box p \vdash \Box q$$

Its tableau (figure 1.2) needs one new world, w_1, and closes on both branches there.

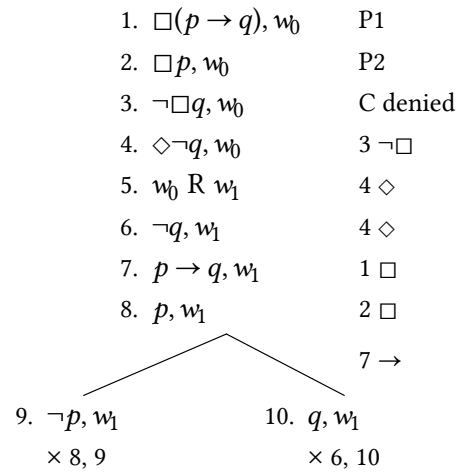


Figure 1.2. Axiom K in the logic K: $\Box(p \to q), \Box p \vdash \Box q$.

1.3 When a branch stays open

What is necessary, one would think, is possible: from $\Box p$, $\Diamond p$. In K it does not follow, and the tableau shows why. Denying $\Diamond p$ gives $\Box\neg p$, but no rule of K draws a world from two boxes, so nothing is left to apply and the branch stays open (↑). An open branch is a recipe for a countermodel: one world that sees no world, where every $\Box$ is true for want of anything to check and every $\Diamond$ is false (figure 1.3, table 1.1). The cure is to require that every world see at least one world. This is the logic KD, and its D is the rule that draws a new world when none is there. Read $\Box$ as "it ought to be that" and $\Diamond$ as "it is permitted that", written O and P, and KD becomes the standard deontic logic. Its axiom is that what is obligatory is permitted, and now the tree closes (figure 1.4).

1.	$\Box p, w_0$	P1
2.	$\neg\Diamond p, w_0$	C denied
3.	$\Box\neg p, w_0$	2 $\neg\Diamond$
	↑	

Figure 1.3. From $\Box p$ to $\Diamond p$ in K: a branch stays open, and the argument is not valid.

	w_0
p	F
R(w)	$\varnothing$

Table 1.1. The countermodel to $\Box p \vdash \Diamond p$ in K: one world, w_0, where p is false and which sees no world.

1.	O p, w_0	P1
2.	$\neg$P p, w_0	C denied
3.	O $\neg p, w_0$	2 $\neg$P
4.	w_0 R w_1	D
5.	p, w_1	1 O
6.	$\neg p, w_1$	3 O
	× 5, 6	

Figure 1.4. From O p to P p in KD: the rule D draws w_1, and the tree closes.